



Science

Stage 8

Paper 1

2025

Cambridge Lower Secondary Progression Test

Mark Scheme

General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antenna but not for anthen (too close to anther).

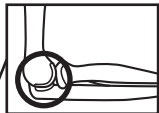
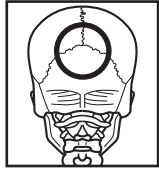
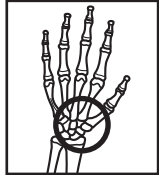
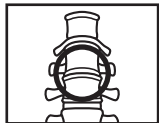
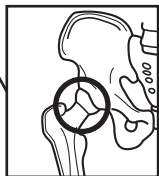
Only the science is being assessed, so answers do not need to be grammatically correct.

Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ or	alternate responses for the same marking point
() brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored.
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

Question	Answer	Marks	Further Information
1(a)	ball and socket joint hinge joint     	2	each correct answer = 1 mark if two lines from one type of joint and one is incorrect = 0 marks for that type of joint
1(b)	antagonistic	1	Accept agonist and antagonist muscles (both needed for the mark) Accept a named pair of antagonistic muscles e.g. biceps and triceps

Question	Answer	Marks	Further Information
2(a)	positive or +	1	Accept plus or +ve
2(b)	negative or –	1	Accept minus or –ve
2(c)	positive charge (on protons) and negative charge (on electrons) (protons and electrons) attract each other	2	each correct answer = 1 mark Accept electrostatic attraction (between protons and electrons) = 2 marks

Question	Answer	Marks	Further Information
3	(speed =) $\frac{\text{distance}}{\text{time}}$ or $\frac{15\,000}{20}$ (speed =) 750 (unit) km/h	3	Accept 750 km/h with working = 3 marks Accept 750 km/h without working = 2 marks Accept 750 without units or working out = 1 mark correct equation or evidence of correct substitution into equation = 1 mark Accept kmph or kilometres per hour Accept speed = $\frac{15\,000\,000}{20}$ 750 000 m/h Accept speed = $\frac{15\,000\,000}{20 \times 60 \times 60}$ 208.33 m/s Note unit must match values used Do not accept ecf from incorrect values for distance and time

Question	Answer	Marks	Further Information
4	(idea that) nail into A has smaller area (idea that) with A there is a greater pressure	2	each correct answer = 1 mark Note assume the answer refers to block A but accept ora if block B specified

Question	Answer	Marks	Further Information
6	any three from mercury is toxic or poisonous bioaccumulation or there is a build-up of toxic chemicals in individuals (bioaccumulation) occurs when mercury cannot be excreted in individuals mercury is found at low levels at the bottom of food chains or much higher concentrations towards the top of food chains/higher stages in a food chain this build-up causes serious nervous system effects not seen at lower stages	3	each correct answer = 1 mark Accept mercury is a toxin Accept bioconcentration for bioaccumulation Accept biomagnification of mercury occurs in the food chain

Question	Answer	Marks	Further Information
7(a)	(idea that) ink will dissolve in the solvent or pencil does not dissolve in the solvent	1	
7(b)(i)	distance to dot A = 37 ± 2 (mm) distance to solvent front = 42 ± 2 (mm)	1	both correct for the mark
7(b)(ii)	0.88	1	Accept ecf from (i) Accept at least two sig figs and truncation
7(c)	red and green and blue	1	all three correct in any order for the mark

Question	Answer	Marks	Further Information
8(a)	The teacher models the Earth, with the core of the Earth behaving as a magnet.	1	more than one answer ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
8(b)	(idea of) Oliver using a compass	1	Do not accept unqualified compass

Question	Answer	Marks	Further Information
9(a)(i)	(question 1) Are there fish swimming in water? (question 2) Does the ecosystem have any trees?	2	each correct answer = 1 mark Note the questions must fit the information in the last column of the key Note the ecosystem has water is not sufficient Accept Does the ecosystem have fish?
9(a)(ii)	(idea of) Are the trees close together?	1	Ignore (idea of) Is the ground around the trees very dark?
9(b)(i)	decreases	1	Ignore they become extinct or they die off
9(b)(ii)	(idea that) farmers who keep animals use evidence to show the harm that eagles cause to their animals or (idea that) farmers who grow fruit use evidence that shows eagles are helpful in reducing crop losses	1	Accept ora Accept ora Accept (idea that) farmers are biased to evidence that suits them

Question	Answer	Marks	Further Information
10(a)	4.0 (g / 100cm ³)	1	Accept 4 (g / 100cm ³)
10(b)	D concentration is the highest	2	each correct answer = 1 mark

Question	Answer	Marks	Further Information
11(a)	36(°)	1	Accept answer in range 34° to 38°
11(b)	angle of incidence = angle of reflection	1	Accept normal and angles marked on diagram with $\angle i = \angle r$ but answer line takes precedence
11(c)	(explanation) (idea that) beam is too wide (improvement) (idea that) narrowing the beam e.g. use a slit or cover the front of the flashlight (torch)	2	each correct answer = 1 mark Accept use a flashlight (torch) with a narrower beam Accept correct answers wherever they are written

Question	Answer	Marks	Further Information
12(a)	biased	1	Accept any indication of the correct answer, e.g. ticking, circling or underlining, but answer line takes precedence
12(b)	look at how global temperatures change over a period of thousands of years	1	more than one answer ticked = 0 marks Accept any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence

Question	Answer	Marks	Further Information
13(a)	glucose + (oxygen) $\longrightarrow$ carbon dioxide + (water)	1	both needed in correct order for mark Ignore sugar
13(b)	mitochondrion	1	Accept mitochondria

Question	Answer	Marks	Further Information
14	no ticked (no mark) any three from most reactive is calcium as it has the lowest time magnesium has the second lowest time and is the next most reactive zinc has the third lowest time and is third in the list of reactivities copper does not react and is therefore the least reactive	3	each correct answer = 1 mark Note for 3 marks answers must quote information from the table Accept order of reactivity should be calcium, magnesium, zinc, copper = 2 marks

Question	Answer	Marks	Further Information
15	(equipment) <u>iron</u> rod power supply and wire (method) (idea that) connect wire to power supply (idea that) wrap wire around the rod	4	each correct answer = 1 mark Accept <u>iron</u> bar for <u>iron</u> rod Do not accept steel rod Accept cell, battery or battery of cells for power supply Accept lead for wire Accept marks from labelled diagram but written answer takes precedence